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Reshaping Reservoir Insights: The Breakthrough Potential of Principal Component-Driven Machine Learning

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Abstract

The oil and gas industry generates a vast volume of logging data during all phases from exploration to production, resulting in an accumulation of high-dimensional logging data. Concluding insightful outcomes from the gathered information is a daunting task due to the multi-dimensionality constraints in data visualization (three axes of space, x, y and z) and data manipulation. The goal is to use a technique to simplify data visualization from high dimensionality into lower dimensionality while preserving features that are important for big data analysis.

One of the most dominant methods to reduce data dimensionality is principal component analysis (PCA), which is used to simplify multivariate datasets acquired by well logging tools (gamma ray, density, neutron porosity, resistivity, etc.) to a smaller number of factors (PCs). The workflow focuses on PCA applications to investigate potentially meaningful petrophysical analysis, reservoir characterization and rock typing, and multi-linear regression analysis for data prediction.

Logging data dimensionality was reduced to produce a set of principal components (PC1, PC2...PCx) where x is equal or close to the number of the used variables. The results indicate that the first two components (PC1 and PC2) represent the majority of the data patterns, and the graphical display of these two components (PCA biplot) shows distinctive clusters that can be assigned to separate electro-facies. Results indicate that PC1 behavior correlates well with lithology variations and can be utilized for well-to-well correlation. Linear regression analysis revealed a strong predictive relevance between PCA components and well logging variables, allowing the R-squared regression technique to predict a result curve from PCA input curves. PC1 showed a strong response to substantial lithological variations, whereas higher principal components are more connected to the formation fluid and could be used to complement the standard petrophysical analysis.

The paper presents an innovative workflow based on PCA components that utilize a range of dimensional reduction approaches to analyze data in a more descriptive form to identify hidden trends and insights as an approach to machine learning and artificial intelligence.

Introduction

Principal Component Analysis (PCA) Background

PCA was invented by [Pearson \(1901\)](#) as an analogue of the principal axis theorem in mechanics. It was later independently developed and named by [Hotelling \(1933\)](#), whilst the best modern references are [Jolliffe \(2002\)](#) and [Niculescu et al. \(2016\)](#). PCA takes a large dataset with many variables and reduces it to a smaller set of summary indices, with the objective of reducing the dimensionality of a large set of variables or features to a smaller one without loss of any important information ([Tharwat 2016](#)).

The computation commences by normalizing each curve. This normalization involves subtracting the mean value of the curve from each data point and subsequently dividing it by the curve's standard deviation. The normalized dataset is then employed in the generation of the principal component curves. This is achieved by performing eigen decomposition on the covariance matrix of the normalized data, which yields the eigenvectors—representing the contribution of each original variable to the principal components. The overall variability within the data for each PC curve can then be calculated accordingly, as exemplified in [Table 1](#) ([Moustafa et al. 2024](#)).

Table 1—A detailed breakdown of the variability and eigen vectors associated with each PC.

	Variability	Eigen vectors			
PC	%	RHOB	DT	NPHI	GR
PC1	48.880	-0.294	0.562	0.666	0.390
PC 2	37.510	0.681	-0.277	0.147	0.660
PC 3	11.060	0.506	0.774	-0.361	-0.117
PC 4	2.540	0.438	-0.085	0.635	-0.630
Input curves normalization					
	Mean Value	2.635	64.537	0.132	79.135
	Std Deviation	0.157	9.541	0.071	43.408

In the above example, the PC1 curve will be computed as follows:

$$PC1 = -0.294 (RHOB - 2.635)/0.157 + 0.562 (DT - 64.537)/9.541 + 0.666 (NPHI - 0.132)/0.071 + 0.390 (GR - 79.135) / 43.408$$

The PCA process produces a series of components, where the first principal component (PC1) accounts for the largest share of variance in the dataset, followed by PC2, and so on. Each component captures a unique dimension of variability, ensuring that the resulting set of principal components remains correlated. This approach removes redundancy present in the original variables, even if they were initially interrelated. By focusing on the leading components that retain most of the variance, it becomes possible to preserve essential information while significantly simplifying the overall data structure.

This streamlined data processing forms a key step in machine-learning workflows by transforming high-dimensional input into a smaller set of principal components that preserve most of the original variance. These components serve as efficient features for machine learning models, helping to improve performance, reduce computational load, and enhance generalization by minimizing noise and multicollinearity.

[Figure 1](#) illustrates the implementation of PCA using proprietary software to process and analyze the dataset utilized in this study. The principal components were generated from a logging-while-drilling (LWD) triple-combo suite, which included measurements from gamma ray, resistivity, density, neutron porosity, and quadrant density logs.

Input		Discriminators	Output	Correlation										
		Use	Default	Log	Well	Well	Well	Well	Well	Well	Well	Well	Well	Well
		Curve	Name		1	2	3	4	5	6	7	8	9	10
Well		→	+		Well Names									
Input Curve 1		→	✓	SGRC	✓	SGRC	SGRC	SGRC	SGRC	SGRC	SGRC	SGRC	SGRC	SGRC
Input Curve 2		→	✓	Density	✓	SBD2	SBD2	SBD2	SBD2	SBD2	SBD2	SBD2	SBD2	SBD2
Input Curve 3		→	✓	Neutron	✓	TNPL	TNPL	TNPL	TNPL	TNPL	TNPL	TNPL	TNPL	TNPL
Input Curve 4		→	✓	Deep_Res	✓	SEDP	SEDP	SEDP	SEDP	SEDP	SEDP	SEDP	SEDP	SEDP
Input Curve 5		→	✓	Shallow_Res	✓	SESP	SESP	SESP	SESP	SESP	SESP	SESP	SESP	SESP
Input Curve 6		→	✓	Medium_Res	✓	SEMP	SEMP	SEMP	SEMP	SEMP	SEMP	SEMP	SEMP	SEMP
Input Curve 7		→	✓	Bottom_Den	✓	ADBQ	ADBQ	ADBQ	ADBQ	ADBQ	ADBQ	ADBQ	ADBQ	ADBQ
Input Curve 8		→	✓	Left_Den	✓	ADLQ	ADLQ	ADLQ	ADLQ	ADLQ	ADLQ	ADLQ	ADLQ	ADLQ
Input Curve 9		→	✓	Right_Den	✓	ADRQ	ADRQ	ADRQ	ADRQ	ADRQ	ADRQ	ADRQ	ADRQ	ADRQ
Input Curve 10		→	✓	Top_Den	✓	ADTQ	ADTQ	ADTQ	ADTQ	ADTQ	ADTQ	ADTQ	ADTQ	ADTQ
Input Curve 11		→												SNP2
Input Curve 12		→												
Input Curve 13		→												
Input Curve 14		→												
Input Curve 15		→												
Input Curve 16		→												
Input Curve 17		→												
Input Curve 18		→												
Input Curve 19		→												
Input Curve 20		→												
Use Well		→		for Model Build		✓	✓	✓	✓	✓	✓	✓	✓	✓

Figure 1—Input LWD data selection matrix for PCA model construction. The LWD curves were from 18 wells, including gamma ray (SGRC), density (SBD2), neutron porosity (TNPL), resistivity (SESP, SEMP and SEDP), and quadrant-based density measurements (ADBQ, ADLQ, ADRQ, ADTQ). The curves were standardized across wells and marked for inclusion in the PCA model build.

Figure 2 presents the explained variability associated with each principal component. The first principal component (PC1) captures 79.3% of the total variance in the dataset, followed by PC2 with 10.9% and PC3 with 5.3%. Collectively, the first three components account for over 95% of the cumulative variability, allowing the original multi-log dataset to be effectively represented by just three components. This substantial dimensionality reduction highlights the strength of PCA in revealing dominant data patterns and relationships across the LWD input variables, enabling more focused interpretation and facilitating advanced reservoir characterization.

Output Curves and Results														
Output	Curve	Variability	Cumulative	Eigen	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector	E. Vector
Curve	Name	%	Variability %	Value	SGRC	sbd2	tnpl	sedp	semp	semp	ADBQ	ADLQ	ADRQ	ADTQ
✓	PrinComp1	79.3	79.3	7.934	0.08	-0.347	0.326	-0.309	-0.324	-0.324	-0.328	-0.346	-0.343	-0.342
✓	PrinComp2	10.9	90.3	1.093	0.885	-0.041	0.197	0.268	0.131	0.216	-0.168	-0.036	-0.033	0.095
✓	PrinComp3	5.3	95.5	0.528	-0.406	-0.225	0.115	0.469	0.367	0.416	-0.343	-0.222	-0.256	-0.112
✓	PrinComp4	1.3	96.9	0.132	0.174	-0.159	-0.905	0.089	-0.058	-0.047	-0.095	-0.216	-0.146	-0.185
✓	PrinComp5	1.1	98	0.112	-0.072	0.068	0.058	0.568	-0.796	0.093	0.119	0.011	-0.061	0.059
✓	PrinComp6	1	98.9	0.096	0.093	-0.018	0.113	0.076	0.128	0.143	0.696	-0.115	-0.068	-0.659
✓	PrinComp7	0.4	99.3	0.037	-0.014	-0.126	0.066	0.508	0.233	-0.726	-0.029	-0.033	0.357	-0.103
✓	PrinComp8	0.3	99.6	0.031	-0.013	-0.241	-0.007	-0.122	-0.188	0.338	-0.247	-0.041	0.787	-0.308
✓	PrinComp9	0.2	99.8	0.021	0.022	0.166	-0.038	0.059	-0.014	-0.038	-0.334	0.769	-0.18	-0.481
✓	PrinComp10	0.2	100	0.017	-0.002	0.834	0.017	0.013	0.01	-0.029	-0.259	-0.42	0.079	-0.231

Figure 2—PCA output summary showing variability and eigenvector contributions. PC1 accounts for approximately 80% of the data variance, with balanced input from all curves. PC2 and PC3 contribute 10.9% and 5.3%, respectively, highlighting specific log influences.

Workflow Overview

Figure 3 illustrates the PCA-driven workflow for analyzing multi-dimensional LWD datasets. Initially, all input log curves are processed through the PCA module, which transforms them into a reduced set of principal components. Among these, the first three components (PC1–PC3) preserve most of the original data variability. These components are then used to perform regression analysis, enabling the prediction of key log measurements. The data is then integrated to identify the formation petrophysical properties to support reservoir interpretation by converting complex, high-dimensional data into a simplified yet informative component set.

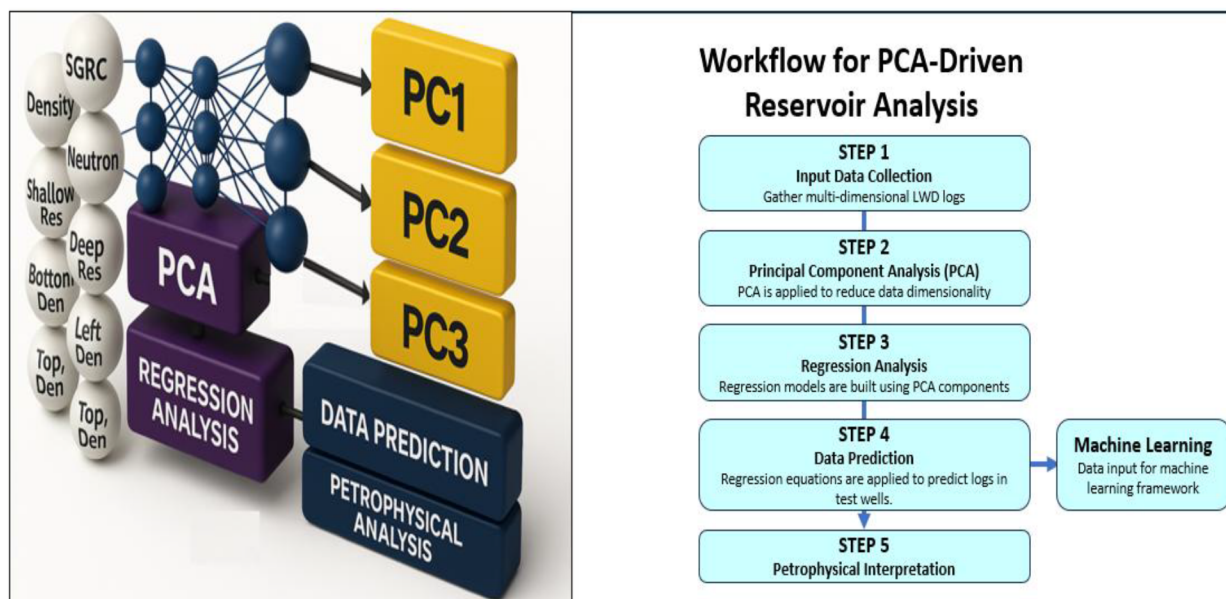


Figure 3—Illustrated workflow for PCA-driven reservoir analysis. The process begins with multi-dimensional LWD inputs transformed into principal components (PC1–PC3) via the PCA model. These components are then used in regression analysis to predict key logs, which are subsequently integrated into petrophysical interpretation.

The workflow started with collecting and preparing LWD data from 18 wells to support the development of a reliable PCA model. Each dataset underwent thorough quality control, including the application of environmental corrections and a detailed review to identify borehole-related or washout effects. Figure 4 illustrates a sample of the real-time LWD data used in this analysis.

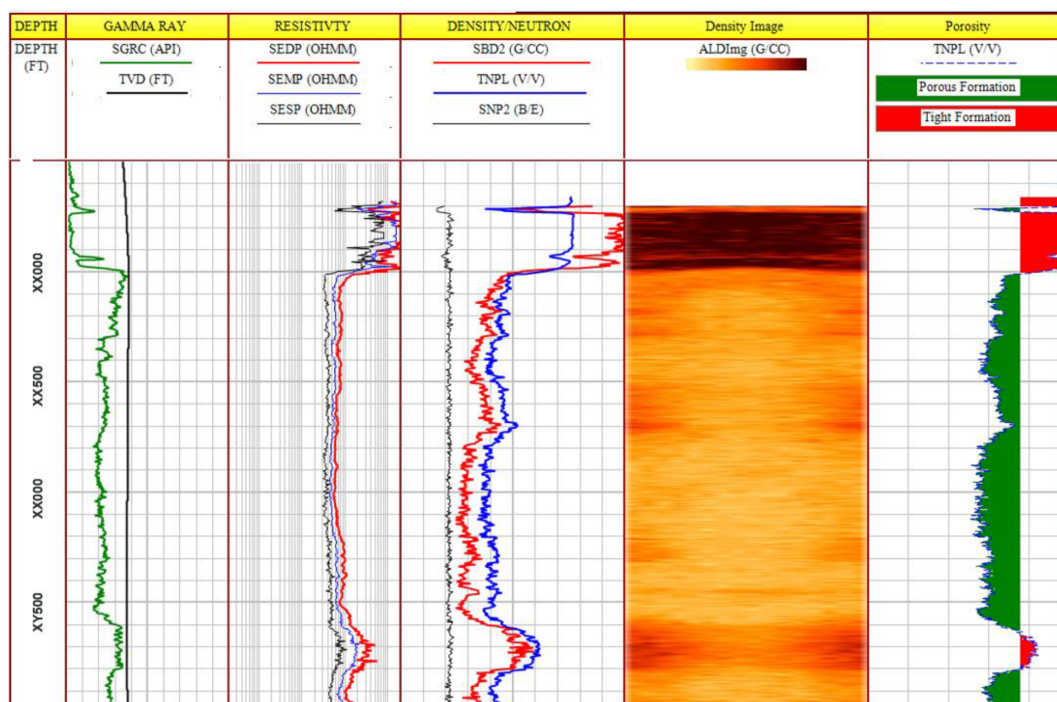


Figure 4—Example of real-time LWD dataset used in PCA modeling. The log suite includes gamma ray, resistivity, density, neutron porosity, and density image logs. The rightmost track classifies formation quality into porous (green) and tight (red) zones, supporting lithology differentiation.

Principal Component Analysis

The PCA model was constructed once the data input was ready. Three key outputs were derived from the model, each of which played a pivotal role in the subsequent detailed analysis. These outputs were as follows:

1. Pearson's correlation coefficients (r). [Figure 5](#) shows a listing of input variables alongside their corresponding correlation coefficient values. These values serve as indicators of the relationships between variables and their magnitude can be evaluated according to the following criteria ([Niculescu et al. 2016](#)):
 - Very high correlation: $r = 0.9\text{--}1.0$
 - High correlation: $r = 0.7\text{--}0.9$
 - Moderate correlation: $r = 0.5\text{--}0.7$
 - Low correlation: $r = 0.3\text{--}0.5$
 - Little or no correlation: $r = 0.0\text{--}0.3$.
2. PCA biplot analysis ([Figure 6](#)). The PCA biplot is a powerful tool for understanding both the distribution of data points and the relationships between variables within the PCA analysis. The biplot has two components:
 - Score plot ([Figure 6](#), right side). This plot displays the distribution of data points in a reduced-dimensional space defined by the principal components. The relative positions of the points reflect similarities or differences in their log responses, allowing for the recognition of patterns, clusters, and trends within the dataset.
 - Loading plot ([Figure 6](#), left side). Within this component, variables are represented as vectors originating from the origin (0,0) of the plot. The direction and length of these vectors signify the extent to which variables contribute to the principal components. Variables closely related, with high correlation coefficients, will be plotted close together, while those with low correlation coefficients will appear farther apart ([ElGhonimy 2015](#)). The relationship can be interpreted as shown in [Table 2](#).
3. PCA quality ([Figure 7](#)) presents the relative contribution of each LWD log to the calculated principal components. It highlights how PC1 integrates significant input from most curves, reflecting a balanced representation of formation properties. In contrast, subsequent components (e.g., PC2 and PC3) are more influenced by specific log responses, such as PC2 being heavily loaded from GR data. This information can be utilized to give insights into distinct petrophysical features such as lithology or fluid content. Together, these outputs form the core of the analytical framework, establishing a solid basis for deeper interpretation and detailed analysis in the following phases of the study.

Table 2—Interpretation of vector orientation in PCA biplot. This table summarizes how the angular relationship between vectors reflects the type and strength of correlation among variables in a PCA loading plot.

Vector Orientation	Correlation Type	Interpretation
Vectors closely aligned	Positive correlation	Variables increase or decrease together
Vectors pointing in opposite directions	Negative correlation	One variable increase as the other decreases
Vectors at right angles ($\approx 90^\circ$)	No correlation	Variables are independent of each other

Pearson's r Correlations

	SGRC	Density	Neutron	Deep_Res	Shallow_Res	Medium_Res	Bottom_Den	Left_Den	Right_Den	Top_Den
Neutron	0.352	-0.899	1	-0.717	-0.783	-0.758	-0.884	-0.891	-0.893	-0.855
Deep_Res	-0.039	0.783	-0.717	1	0.879	0.951	0.681	0.78	0.766	0.834
Shallow_Res	-0.151	0.838	-0.783	0.879	1	0.93	0.753	0.841	0.834	0.861
Medium_Res	-0.086	0.833	-0.758	0.951	0.93	1	0.737	0.831	0.816	0.869
Bottom_Den	-0.295	0.948	-0.884	0.681	0.753	0.737	1	0.938	0.937	0.859
Left_Den	-0.212	0.981	-0.891	0.78	0.841	0.831	0.938	1	0.974	0.955
Right_Den	-0.2	0.972	-0.893	0.766	0.834	0.816	0.937	0.974	1	0.944

Figure 5—Pearson's correlation coefficients (r) used to quantify the strength and direction of a relationship between two continuous variables. These correlations assess how closely two variables are related and whether they tend to increase or decrease together. The Pearson correlation coefficient produces a value between -1 (strong negative correlation) and 1 (strong positive correlation).

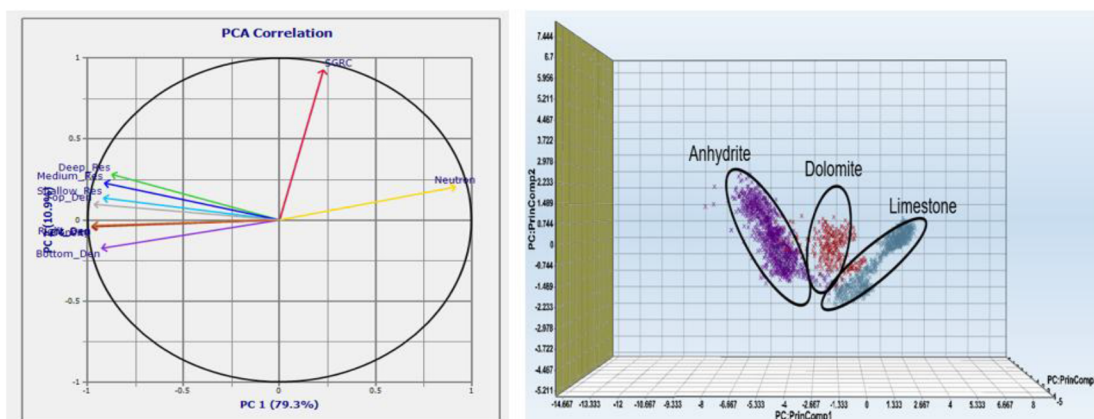


Figure 6—Left Side: Loading plot displaying the relationships between variables. Right Side: Scoring plot displaying the scoring values of data points, which helps to visualize how data points cluster or spread out in the reduced-dimensional space defined by the selected PCs.

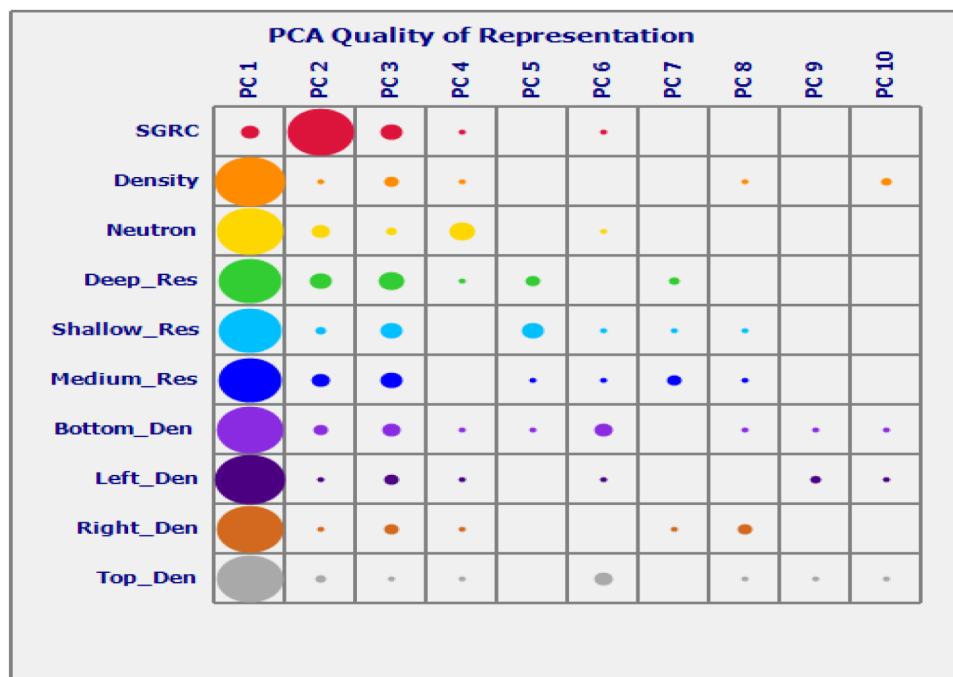


Figure 7—PCA quality of representation. This bubble chart highlights the relative contribution of each input log to the computed principal components. PC1 captures balanced input from most variables, while PC2 shows a dominant influence from SGRC, reflecting differentiated sensitivities across components.

Regression Analysis Using PCA Data

This section focuses on key analytical insights: correlation analysis, regression analysis, and data prediction.

Correlation Analysis

Before proceeding to the regression stage, it was essential to examine the interrelationships among the input LWD variables. The PCA loading plot in Figure 8 provides a visual representation of these correlations through the orientation and length of vectors. The plot reveals a clear spatial separation between variables and two clear groups of data. SGRC and Neutron vectors are located on the right half of the plot. In contrast, vectors positioned on the left half including resistivity and density measurements are oriented in the opposite direction, indicating a negative correlation with SGRC and Neutron. The relative angles and positioning of the vectors suggest the following relationships:

- SGRC and Neutron logs are positively correlated, as shown by their aligned directions on the right, however the correlation is weak.
- Resistivity and density logs are positively correlated among themselves but negatively correlated with SGRC and Neutron.
- Logs at near-perpendicular angles, such as SGRC and some shallow density measurements, show weak or no correlation.

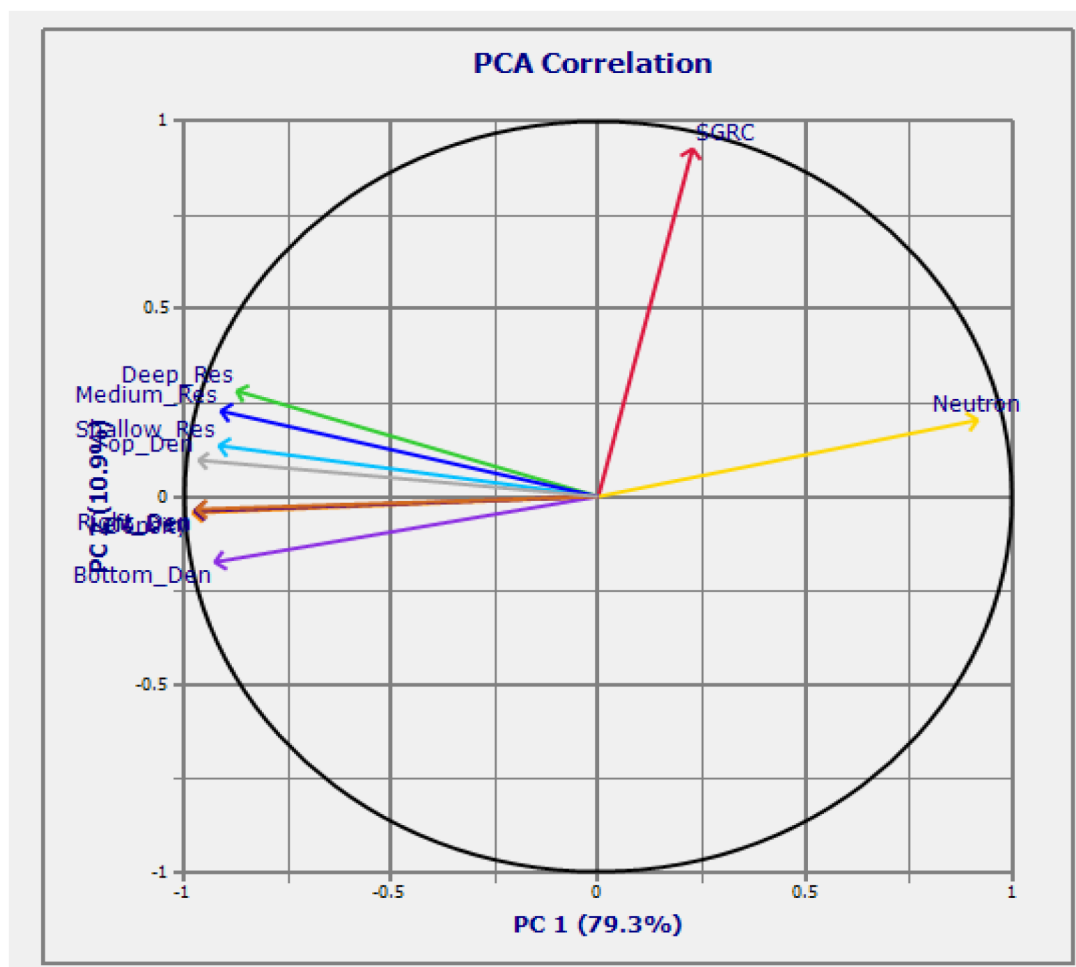


Figure 8—PCA The PCA loading plot shows how the input logs relate to each other. SGRC and Neutron point in the same (positive) direction, while resistivity and density point in the opposite direction, clearly separating the variables for easier analysis., indicating clear variable separation for effective multivariate analysis.

Regression Analysis

Following the establishment of the PCA model and considering the strong correlations between the inputs identified in the previous section, regression analysis was the next step. The analysis was conducted using proprietary software, focusing on two distinct regressions. The first regression aimed to predict porosity from PC1, characterized by significant loadings from various variables. Another regression was constructed to predict the density data using the relationship with PC1. The primary objective was to utilize the training data from 18 wells, which was used in constructing the PCA model, to predict both porosity and density.

The process can be broken down into three key steps as illustrated in Figure 9.

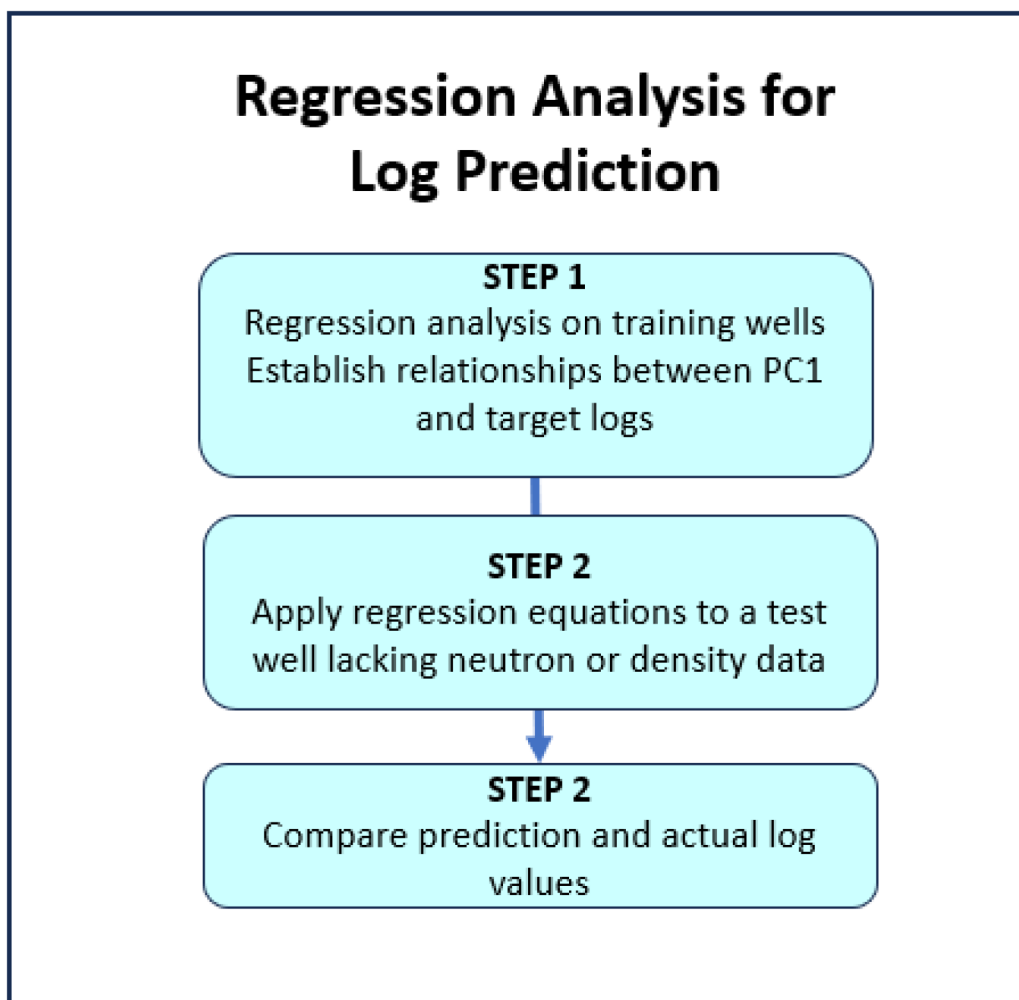


Figure 9—Regression analysis workflow. A three-step process for predicting target logs from PCA data: (1) regression modeling using training wells, (2) application of regression equations to a test well, and (3) validation by comparing predicted versus actual log responses.

Data Prediction - Density

Figure 10 displays a cross-plot of PC1 versus density using data from 18 wells across different rock types. Several regression models—including linear, polynomial, and exponential were applied to capture the relationship between PC1 and density. The plot confirms that a linear fit alone ($R^2 = 0.8309$) is insufficient to represent the data trend accurately. Instead, polynomial ($R^2 = 0.8654$) and exponential ($R^2 = 0.8923$) regressions provided a better fit, as reflected by the corresponding equations shown in Figure 11.

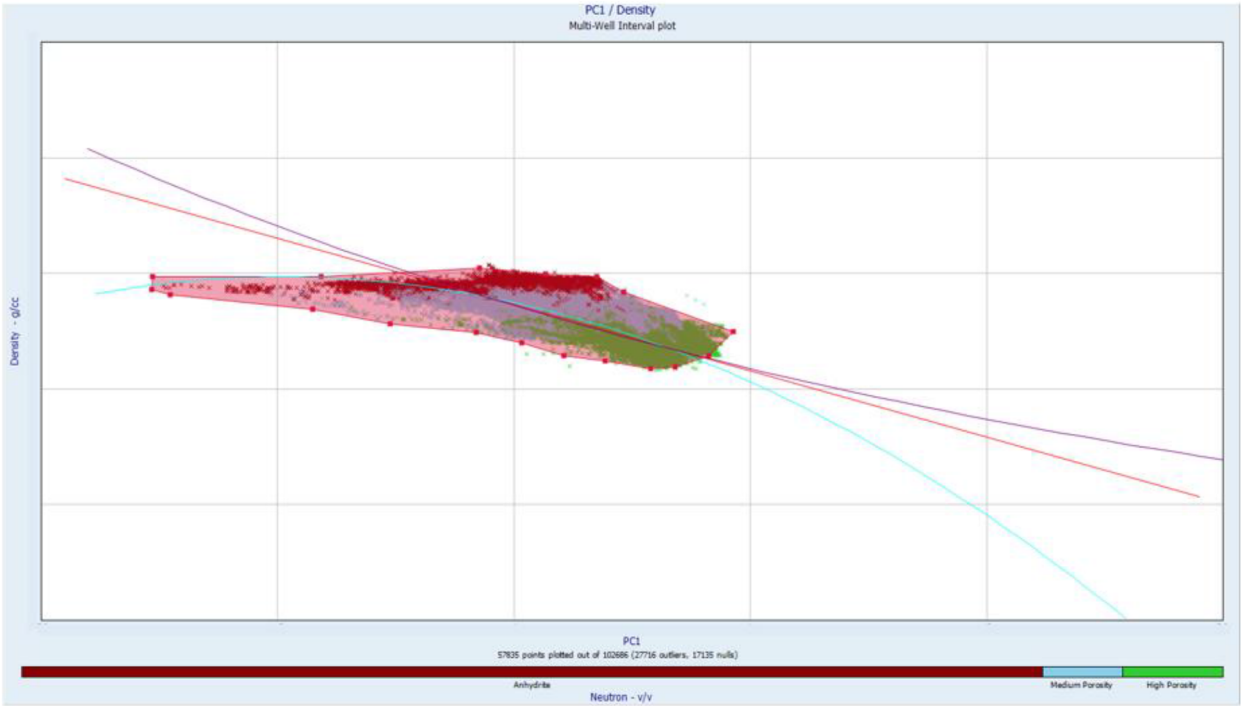


Figure 10—PC1 vs. density cross-plot. Multivariate cross-plot from 18 training wells showing the relationship between PC1 and bulk density across various lithologies. Linear, polynomial, and exponential regression curves are overlaid to evaluate fit quality. The curved trend confirms the need for nonlinear regression methods.

Multi-Line Formulae - UTMN_1092

Line	Use	Cont.	Out	Name	Type	Array	Unit	Formula
1	✓		✓	Density_ALL_EXP	Numeric		g/cc	2.52783658 * exp (-0.0285518821 * PC:PrinComp1)
2	✓		✓	Density_ALL_P	Numeric		g/cc	2.58988329 - 0.0857462895 * PC:PrinComp1 - 0.00334043071 * PC:PrinComp1 **2
3	✓		✓	Density_ALL_L	Numeric		g/cc	2.50596442 - 0.0727694947 * PC:PrinComp1

Figure 11—Regression equations used for density prediction. Regression formulas developed from PC1 using training well data. Three models are presented: exponential (Density_ALL_EXP), polynomial (Density_ALL_P), and linear (Density_ALL_L). Each captures the relationship between PC1 and density with varying levels of curvature to match formation characteristics.

Figure 12 presents a comparison between the actual density log from Well B and the predicted density curves generated using three different field-based regression models: linear (Density_ALL_L), polynomial (Density_ALL_P), and exponential (Density_ALL_EXP). Overall, exponential regression shows a strong agreement with the measured density across the carbonate interval. However, noticeable mismatches appear within the tight formation zones, indicating that a single regression model does not adequately capture the complexity of these intervals. This discrepancy indicates the need for zone-specific regression models tailored to distinct lithological layers.

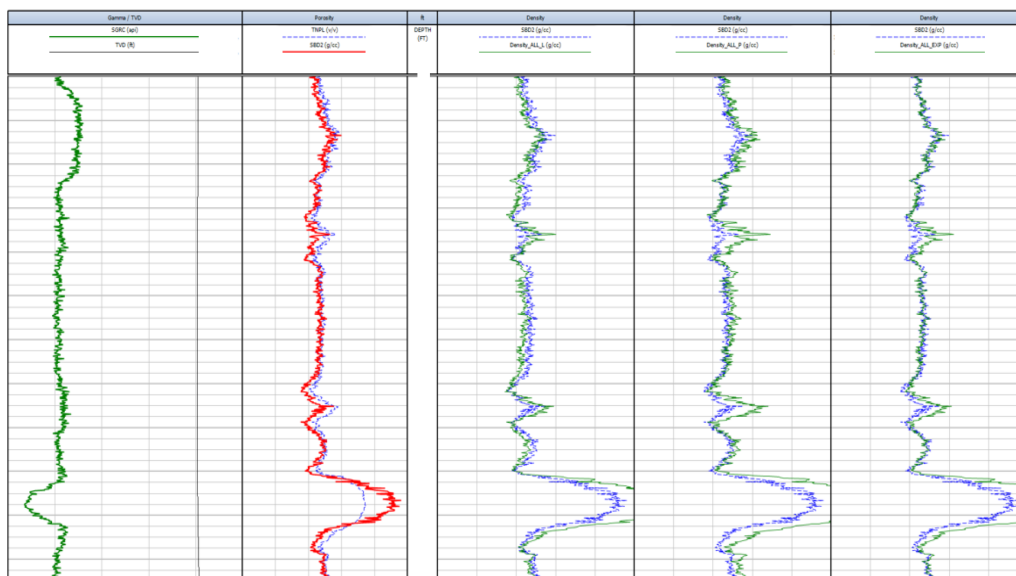


Figure 12—Comparison between actual and predicted density logs for Well B. The predicted curves are compared to the original density log. The exponential model shows the closest alignment in the carbonate interval, while mismatches in tight zones suggest the need for zone-specific regression models.

To further validate the accuracy and reliability of the field-derived PCA data, PC1 was applied to a single test well B, as shown in Figure 13. This step aimed to assess the model's predictive capability across different reservoir qualities, including both porous and tight formations. The cross-plot of PC1 versus density from Well B was used to generate a predicted density curve based on three regression models. The resulting curves (Figure 14) show a strong match with the actual density log across the porous limestone interval. Moreover, the model maintained reasonable consistency even within tight formations, indicating that the PCA-based regression approach can effectively capture both high- and low-porosity lithologies using a single principal component. This outcome supports the model's applicability for field-scale log prediction.

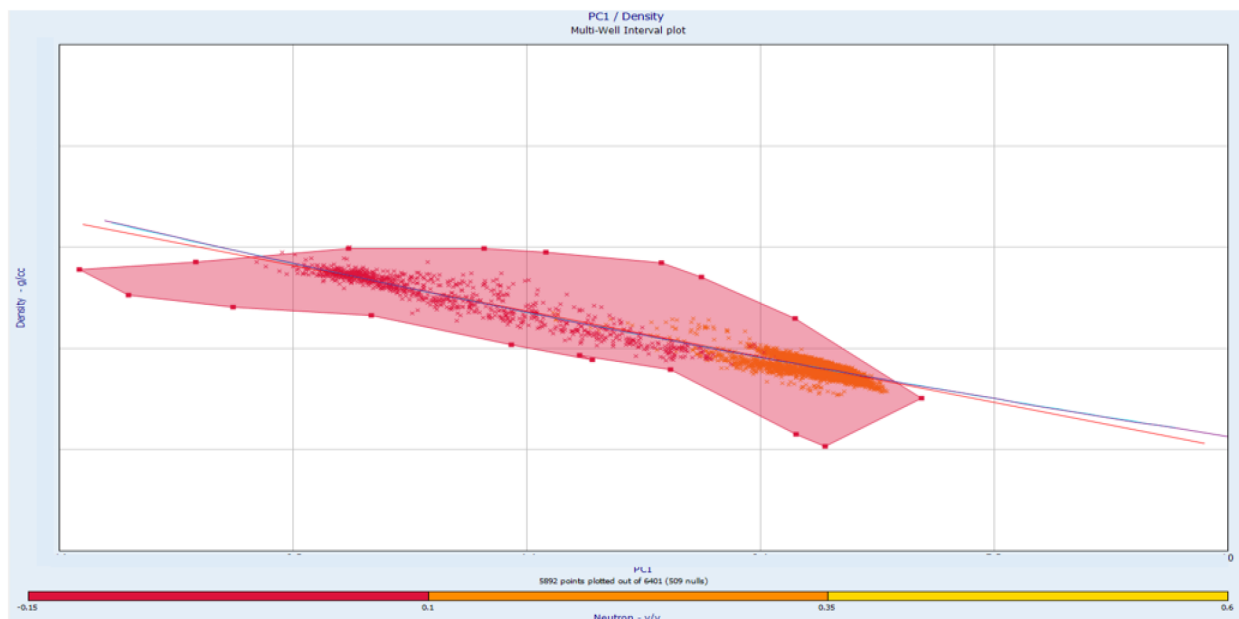


Figure 13—Validation cross-plot of PC1 versus density for a single test well.

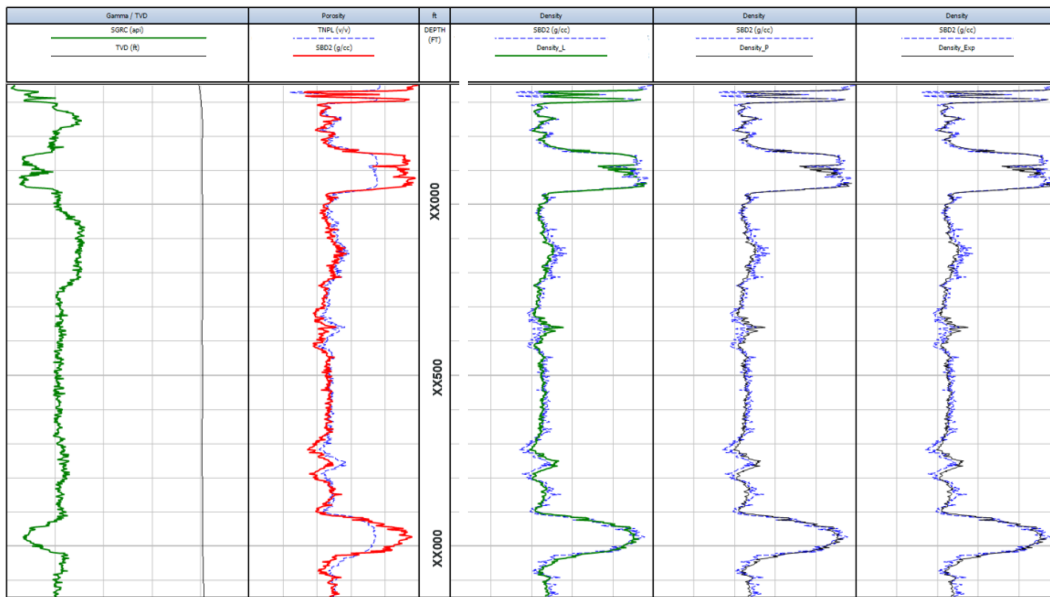


Figure 14—Comparison between predicted and actual density logs using PCA-based regression. It demonstrates strong agreement across the porous limestone interval and maintains consistency in tight formations.

Data Prediction – Neutron Porosity

The same methodology used for density prediction was applied to estimate neutron porosity. Figure 15 shows a cross-plot of PC1 versus neutron porosity from the training wells. Based on this, three regression models, TNPL_R_Linear, TNPL_R_Poly, and TNPL_R_Exp, were developed and applied to a test well. As illustrated in Figure 16, the exponential regression model provides the closest match to the actual neutron porosity log, particularly across the carbonate interval, indicating its effectiveness in capturing the underlying trend.

- $TNPL = 0.146 + 0.025 * PC1$
- $TNPL = 0.139 + 0.030 * PC1 + 0.001 * (PC1)^2$
- $TNPL = 0.132 * e^{(0.311 * PC1)}$

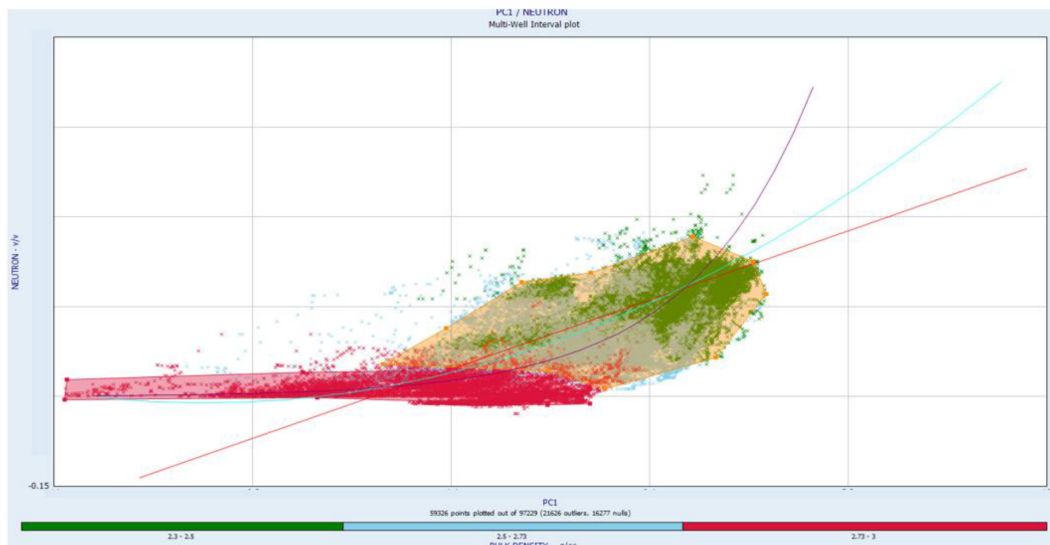


Figure 15—Cross-plot for PC1 vs. neutron porosity curve from training wells exhibits a curved relationship.

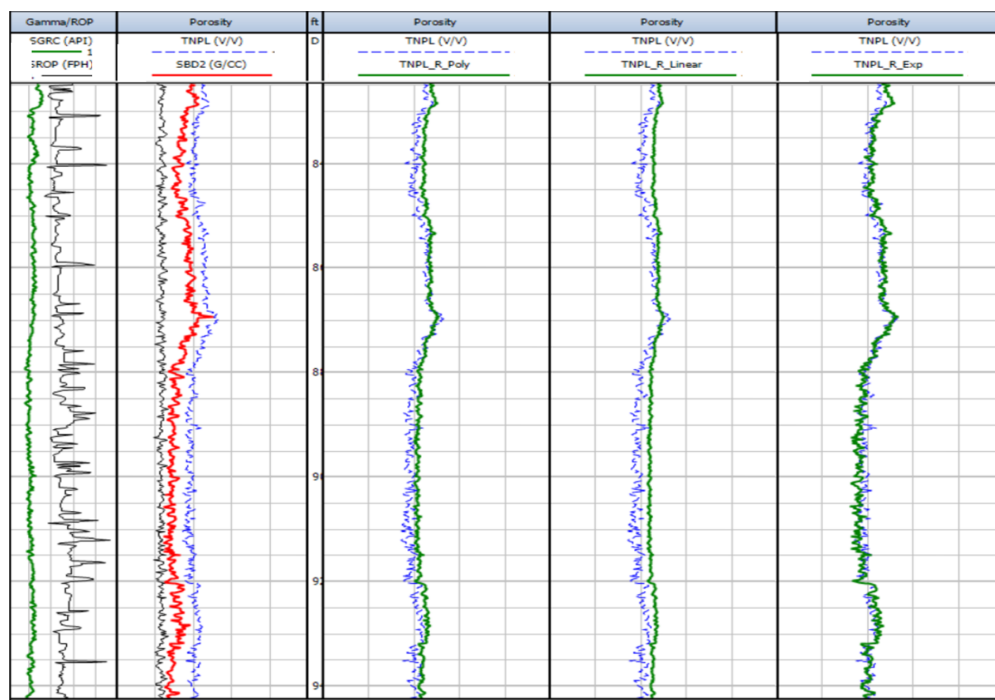


Figure 16—Predicted neutron porosity curves from three regressions (polynomial, linear, and exponential) in green, compared to tested Well-B neutron porosity data in blue. Strong alignment with the exponential regression demonstrates high model accuracy.

Zonal Regression Modeling

To improve prediction accuracy across different lithologies, a zone-specific regression approach was implemented. Figure 17 illustrates two distinct regression models developed based on PC1 (field scale) vs test-well neutron data, showing a porous carbonate interval on the right and a tight formation on the left. These models account for the differing relationships between PC1 and neutron response within each lithological zone. The regression formulas were applied conditionally using a density discriminator, ensuring that each model was activated only within its appropriate rock type.

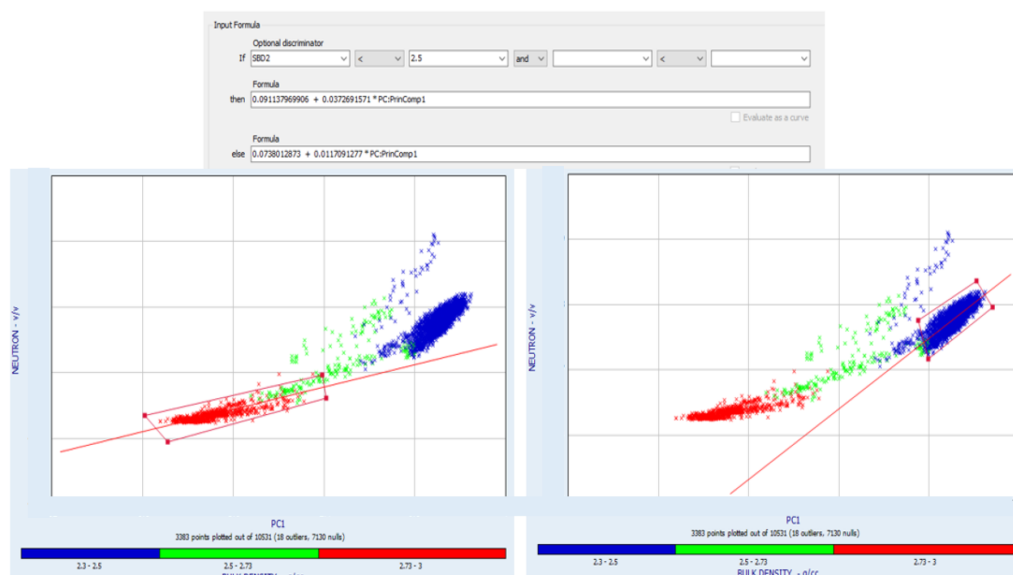


Figure 17—Zonal regression for neutron prediction. Two regression models were applied conditionally using density-based zonation. The right-hand cross-plot demonstrates the porous formation while the left side is for the tight formation.

In Figure 18, the difference in performance between the two approaches is clearly observed. The curve labeled TNPL_R_Lim represents the predicted neutron porosity using a single field-based regression equation; it does not take into consideration different lithologies. This curve shows a visible mismatch across the tight interval, where the model fails to capture the lower porosity trend. In contrast, TNPL_R_Lim_An timer incorporates zone-specific regression, conditionally applied based on lithology. It shows significantly improved alignment with the measured neutron porosity curve across both porous and tight intervals. This demonstrates the effectiveness of zonal regression in capturing variable formation responses and highly recommended to be utilized to reduce prediction error in heterogeneous reservoirs.

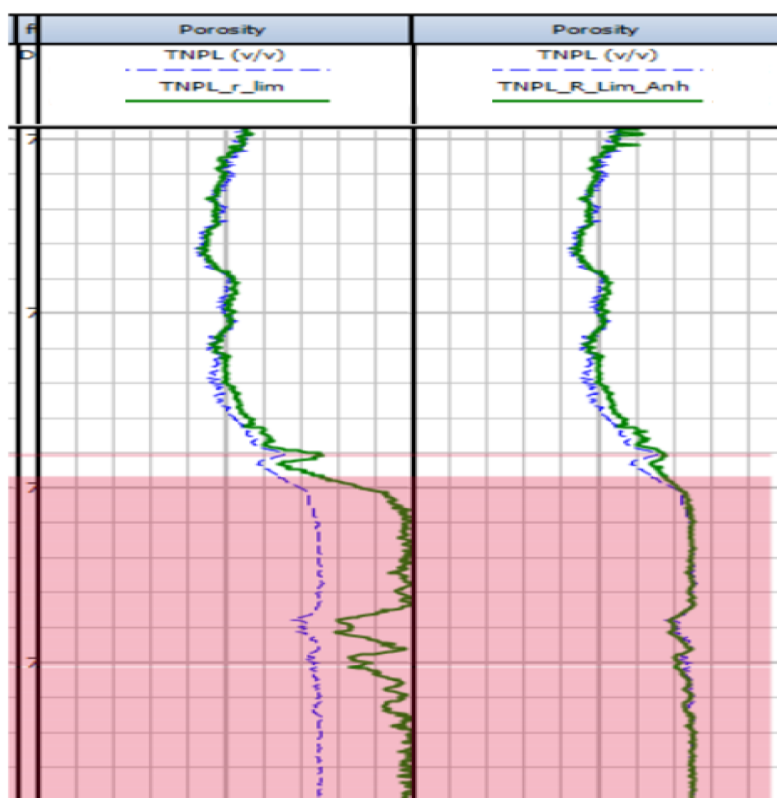


Figure 18—Left: TNPL_R_Lim from field-wide regression shows mismatch in tight formation. Right: TNPL_R_Lim_An timer from zonal regression improves accuracy by accounting for lithology-specific behavior.

PCA-Based Petrophysical Analysis

PCA, when compared to petrophysical analysis, can be particularly enlightening, especially when we examine the first two principal components, PC1 and PC2. PC1 often captures the most significant variations in the data, and in the context of petrophysics, it may reveal the dominant geological factors influencing the dataset (Moustafa et al. 2024). For instance, high loading of resistivity and gamma-ray measurements on PC1 could suggest variations in lithology or formation fluid content. PC2, in turn, tends to unveil the second most substantial source of variance. In petrophysics, a high loading of parameters like porosity and density in addition to the resistivity on PC2 may signify the presence of distinct rock types or stratigraphic layers. To illustrate this further, below are two cases—one for clastic reservoirs and the other for carbonate reservoirs—where PCA, when correlated with conventional logs, provides invaluable insights into the fundamental geological attributes entrenched within subsurface formations. Figure 19 shows an example from a clastic reservoir drilled in a horizontal section. The log shows a comparison between the conventional log interpretation results and the PCA components. The PCA results reveal that PC1 accounts for a substantial portion (84.9%) of the overall variance in the dataset (see Figure 20). PC1 exhibits uniform

loadings for all the logs. It plays a pivotal role as a primary lithological discriminator. The results show PC1 effectively segregating the upper shale interval, characterized by positive PC values and negligible effective porosity, from the high-porosity sand formation, characterized by negative PC values in the lower section. The comparison underscores a strong correlation between PC1 and effective porosity, accurately distinguishing the reservoir boundaries.

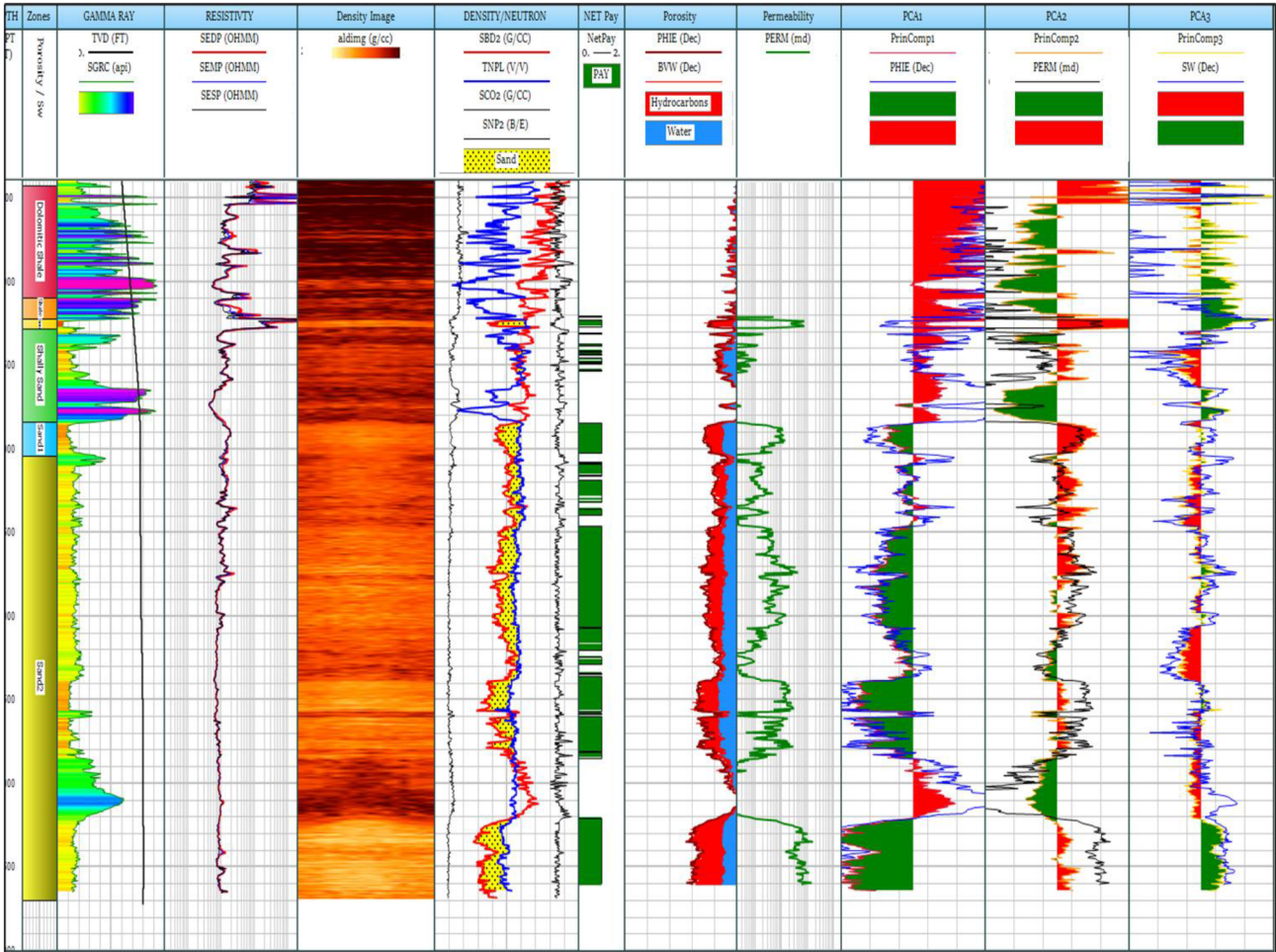


Figure 19—A PCA components vs. petrophysical interpretation in a clastic reservoir. PC1 aligns with effective porosity (PHIE) trend, PC2 with permeability, and PC3 with water saturation (SW), demonstrating the value of PCA for lithology and formation properties evaluation.

Output Curve Set : PC (Principal Component)													
Principal component analysis run on model build wells													
Results for all wells in Model build													
Variability, PCA Eigen results													
PC	% Variability	% Cum Vari	Eigen Value	SGRC	SB02	AD0Q	ADLQ	AD0Q	ADTQ	SEDP	SEMP	SESP	TNPL
1st	84.97	84.97	10.19677	0.24873	-0.30649	-0.27996	-0.30195	-0.30377	-0.28378	-0.27934	-0.27683	-0.28903	0.28649
2nd	6.95	91.93	0.83436	-0.41907	-0.15592	-0.3339	-0.16889	-0.15352	0.06509	0.43928	0.45014	0.16749	0.29624
3rd	2.87	94.8	0.3446	0.77233	-0.03138	-0.25283	0.00605	-0.02443	0.26779	0.20123	0.24018	0.16332	0.27199

Figure 20—Variability for each PCA component, indicating that PC1 involves most of the variability and captures most of the data.

The second principal component, PC2 contributes to approximately 7% of the total variance within the log suite. It is strongly influenced by the integration of resistivity, gamma ray and porosity data. PC2 serves as an effective separator, distinguishing porous and permeable intervals, irrespective of their lithological composition (indicated by positive score values), from impermeable formations (indicated by negative score values).

values), encompassing shales or very low permeability layers. Comparisons with permeability curves reveal a notably strong correlation and alignment.

Higher-order components such as PC3 exhibit greater sensitivity to fluid content. PC3 accounts for approximately 2.87% of the total data variability that is independent of PC1 and PC2. Notably, it demonstrates heightened loadings for resistivity curves (SESP, SEMP, and SEDP), indicating a distinct response to fluid properties. This is corroborated by a strong correlation between PC3 and water saturation, underscoring its utility in fluid-related interpretations.

Figure 21 presents an additional case in carbonate, yielding results consistent with those observed in the previous example. In this scenario, PC1 continues to exhibit a robust association with effective porosity, further solidifying its role as a crucial discriminator, effectively delineating the target reservoir zone from non-reservoir formations. PC2 shows a good alignment with the permeability curve, underscoring its significance in understanding the formation's permeability characteristics. These findings emphasize the consistency of PCA's performance in different geological contexts and its capacity to deliver valuable insights for reservoir characterization.

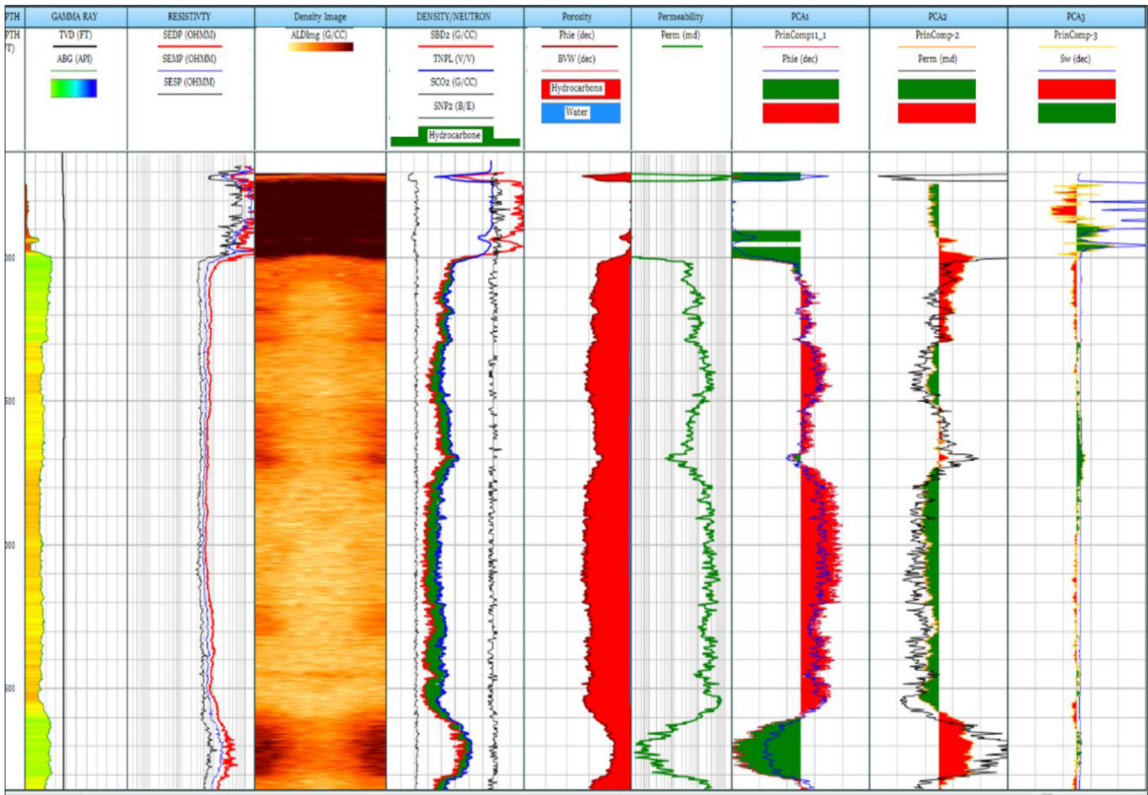


Figure 21—PCA components vs. petrophysical interpretation in a carbonate reservoir. PC1 continues to correlate with effective porosity (PHIE), PC2 tracks permeability trends.

Conclusion

This study demonstrated the value of PCA as a powerful tool for simplifying and extracting insights from multi-dimensional LWD datasets. PCA successfully reduced data complexity while preserving critical geological and petrophysical information. The first principal component (PC1) showed strong correlation with lithology, enabling effective reservoir characterization and data-driven well-to-well correlation.

Building on the performed regression and data prediction insights, PCA has demonstrated its ability to distill complex, high-dimensional log data into a concise set of principal components, making it a powerful tool for field-scale data prediction. These components can be further utilized in later stages as input

features for machine learning workflows. By capturing key patterns and formation variability, reducing noise and preserving geological relevance, PCA improves model accuracy, accelerates training, and showed the potential for enhancing the generalization of machine learning models in reservoir characterization.

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